

AI-Driven Dynamic Pricing for High-Value Assets in Manufacturing and Services: Optimizing Finite Horizon Sales with Demand Sensitivity

Meilan Chen^a, Xiangling Hu^b, Yuan Qi^{c*}, Donato Masi^d

^aSchool of International Business, Guangdong University of Finance & Economics, Foshan, China;

meilanchen@gdufe.edu.cn

^bSeidman College of Business, Grand Valley State University, USA; hux@gvsu.edu

^cSchool of Business Administration, Shanghai Lixin University of Accounting and Finance, Shanghai, China;

nikkeshu@126.com

^dAston Business School, Aston University, Birmingham, UK; d.masi@aston.ac.uk

Abstract

In the context of AI-driven manufacturing and service industries, the strategic selling of high-value products within a finite time horizon is a critical challenge for maximizing expected profit. This research investigates how AI can be leveraged to enhance dynamic pricing strategies, where historical prices influence each customer's offer. Employing AI algorithms, the seller dynamically adjusts the minimum acceptable prices at various time points, responding to market trends and predictive analytics. Our study reveals that in scenarios where AI anticipates an increasing trend in offered prices, sellers are inclined to delay sales to capitalize on potentially higher future offers. Conversely, in situations where AI predicts a decreasing trend in offered prices, the algorithm adjusts the minimum acceptable price to be an increasing function of the remaining sales time, optimizing the timing of sales for individual product units. Additionally, when dealing with two distinct products, the AI-driven pricing strategy adapts the minimum acceptable prices based on the relative cost magnitudes of these products. This research underscores the potential of AI in transforming traditional dynamic pricing approaches, offering novel insights into how AI-enabled tools can optimize sales strategies in the manufacturing and service sectors, balancing profitability with market responsiveness.

Keywords: Dynamic pricing; Finite time horizon; AI-enabled manufacturing; Industrial product

1 Introduction

The swift progress in artificial intelligence (AI) is driving transformative changes in the manufacturing and services industries. In manufacturing, AI-powered robotics are revolutionizing production lines by enhancing efficiency, reducing human error, and enabling the customization of products at scale (Marr, 2020). In the services sector, AI is equally transformative, with chatbots and virtual assistants providing enhanced customer service, personalizing interactions based on customer data, and streamlining operations (Link et al., 2020). AI-driven analytics are also enabling companies to predict consumer trends, optimize inventory management, and improve decision-making processes. These advancements are not only optimizing operational efficiencies but also creating new business models and opportunities for innovation across industries (Reim et al., 2020). In this evolving landscape, firms are trying to create synergies and enhanced coordination between different functional areas (Chien et al., 2020; de Giovanni, 2021; Khosrowabadi et al., 2022; Sharma et al., 2022; Xu et al., 2023; Castane et al., 2023) and different stakeholders.

Pricing strategies represent a pivotal mechanism to achieve integration between manufacturing and services, namely between the supply chain, marketing, and finance functional areas. The so-called "pricing/inventory" prices are set in conjunction with inventory decisions or are established according to existing inventory levels (Hall et al., 2009). The evolution of AI has opened new avenues for optimizing pricing strategies within a finite time horizon, and dynamic pricing strategies have shown to be successful across a broad range of sectors such as airlines, hospitality, retail, e-commerce, mobile communication, automotive, sporting events, car rentals, and insurance (Akilandeewari et al., 2024). In the airline sector, this approach is commonly known as yield or revenue management. At the same time, in the automotive industry, dynamic pricing integrates production planning with inventory management to enhance supply chain efficiency and increase profits.

Among the sectors embracing this change, the ones dealing with high-value assets such as industrial machinery, advanced technological equipment, and comprehensive service contracts face unique challenges and opportunities in pricing strategies. In the context of AI-driven markets, sellers of high-value manufacturing or service products can access and analyze vast amounts of historical pricing data. This capability transforms the traditional pricing approach, allowing for more sophisticated and responsive strategies. For instance, in the manufacturing sector, AI can

predict market trends for specialized machinery, enabling manufacturers to adjust prices dynamically based on anticipated demand shifts. Similarly, in the service industry, providers can utilize AI to analyze past contract prices and adjust their offers for future deals.

In light of this opportunity, this paper explores how AI can enhance dynamic pricing decisions by leveraging historical data and predictive analytics. More specifically, this research investigates a scenario where a seller, empowered by AI tools, does not set a fixed price but instead uses AI algorithms to determine the minimum acceptable prices. These prices are dynamically adjusted based on historical sales data, remaining sales time, and inventory levels. When a customer makes an offer, the AI system evaluates it against the current minimum acceptable price. The sale proceeds if the offer meets or exceeds this threshold. This approach allows for maximizing expected profits while adapting to market dynamics.

This AI-enabled pricing strategy implies an enhanced integration between the firm's supply chain, marketing, and finance functional areas. By harnessing comprehensive data, including inventory levels, customer demand, and financial metrics, AI algorithms can optimize pricing decisions in real-time, ensuring they align with the firm's strategic objectives and market conditions. This cross-functional coordination allows for a swift adaptation to supply chain disruptions, shifts in consumer behavior, and financial imperatives, ensuring that pricing strategies not only meet consumer demands but also enhance profitability and competitive advantage. Furthermore, this integrated approach facilitates a personalized customer experience, aligning pricing with customer expectations and supply capabilities, thereby fostering customer satisfaction and loyalty.

The remainder of this paper is structured as follows: Section 2 reviews relevant literature on AI-enabled dynamic pricing in manufacturing and services; Section 3 details the AI-based models and their settings. Section 4 presents an analysis of optimal policies for single and multiple-product scenarios in both manufacturing and service contexts. Section 5 extends the model to scenarios with varying costs and complexities. Section 6 concludes with insights into the future of AI-driven dynamic pricing strategies and potential research directions.

2 Literature review

This paper situates itself at the intersection of AI-driven dynamic pricing and customer behavior analysis, mainly focusing on high-value assets in manufacturing and services. The literature

relevant to our study spans three primary domains: customer behavior in pricing, and dynamic pricing in multi-period settings with independent and dependent demands.

The complexity of customer decision-making processes, especially in the context of high-value asset pricing, has been a subject of study for decades. Wilson et al. (1996) highlighted the intricacies of anchoring and adjustment in judgment processes, which are critical in understanding customer behavior in dynamic pricing scenarios. Wansink et al. (1998) suggested that customers are likely to engage with prices only when adjustments exceed certain thresholds. This concept aligns with the idea of willingness-to-pay (Li et al., 2021) and willingness-to-sell (Pan et al., 2014), both of which are pivotal in AI-driven dynamic pricing models. Our study incorporates the behavioral insights from those customer behavior studies into AI-driven models, ensuring that dynamic pricing strategies are responsive to the identified thresholds and behaviors.

In the realm of dynamic pricing, the foundational work of Bitran and Mondschein (1993, 1997) is particularly relevant. They explored multi-period pricing strategies, emphasizing the role of inventory and time in pricing decisions. Their findings about non-increasing optimal prices over time and inventory levels provide a basis for understanding how AI algorithms might approach dynamic pricing in a similar context. Zhao and Zheng (2000) and Feng and Xiao (2000) further expanded on these concepts, exploring the properties of optimal pricing in relation to inventory levels and time under varying conditions of customer demand and price sets. Our study further extends these models into a more dynamic, real-time context facilitated by AI. It builds on their findings by integrating AI algorithms to dynamically adjust pricing based on historical sales data, remaining sales time, and inventory levels.

The role of imperfect demand information, a key aspect in AI-driven models, has been explored by researchers like Araman and Caldentey (2009), Moon and Silver (2000), and Besbes and Sauré (2014). Their work on learning processes and heuristic development for optimal pricing policies under imperfect information conditions is directly applicable to AI models that must navigate similar uncertainties.

The dependency of new demand on past prices, a critical aspect of our study, has been addressed in studies like Petruzzi and Monahan (2003), Ahn et al. (2007), and Güler et al. (2014). These studies provide insights into how historical pricing data can be leveraged in AI models to predict future demand and set optimal prices. Chen et al. (2016) further contributed to this field by

developing algorithms for dynamic pricing considering reference prices, a concept that can be integrated into AI-based pricing strategies. Our research extends this work by employing AI to analyze vast amounts of historical data so as to design more precise and adaptive pricing strategies to account for the dependencies of new demand on past prices.

The broader field of dynamic pricing has seen contributions from various researchers. Chatwin (2000) explored the implications of restocking in dynamic pricing models, while Vulcano et al. (2002) investigated auction mechanisms in this context. Su (2007), Swami and Khairnar (2006), and Liu and van Ryzin (2008) provide diverse perspectives on customer behavior, pricing, and inventory strategies, all of which are relevant to AI-driven models. Gallego and Hu (2014) and Feng and Gallego (1995) offered insights into price competition and strategic price changes in the presence of perishable assets. This scenario can be mirrored in AI models for high-value assets in manufacturing and services and other fields like real estate and finances. Recent discussions by Seele et al. (2021) and Sharma et al. (2022) on the ethical implications of AI in pricing highlight the need for transparency and fairness.

Finally, the specific case of high-value asset sales, such as real estate in Canada, as discussed by Meredith and Akinc (2012), provides a direct parallel to our study's focus. Their findings on optimal pricing in the context of finite time horizons and single-unit sales offer a valuable perspective for AI-driven dynamic pricing models in similar high-value asset scenarios.

Overall, this paper contributes to the existing literature by integrating AI into dynamic pricing models, addressing the complexities of customer behavior, imperfect demand information, and historical data utilization. It also considers ethical implications and provides real-world applications, and therefore provides a holistic approach to AI-driven dynamic pricing strategies for high-value assets.

3 Model

In the context of an experimental launch of an AI-related product or service, the model adapts to optimize pricing strategies for selling innovative AI solutions within a finite time frame. This model is particularly relevant for industries exploring cutting-edge AI technologies, such as autonomous machinery, AI-driven healthcare devices, or personalized AI-based service platforms.

The integration of AI in this model not only enhances the pricing strategy but also reflects the nature of the product being sold.

The seller's goal is to maximize the total expected profit by selling N units of a novel AI product or service within a predetermined time frame. The model focuses on scenarios where $N=1$ or $N=2$, indicative of the product's exclusivity and high developmental cost. An example could be a limited release of an advanced AI-powered medical diagnostic tool or a bespoke AI-driven personal finance advisor platform.

The selling period is divided into T time intervals, with one customer arriving in each interval. This structure allows for a detailed analysis of customer interactions and pricing strategy adjustments over time, crucial for a product in its experimental launch phase. Suppose that the random change rates at different intervals are always positive, independently and identically distributed with a certain distribution F with a finite mean μ_F . The total cost, including development, marketing, and production costs, is denoted as c per product. Given the innovative nature of the product, the salvage value of unsold units at the end of the time horizon is assumed to be zero, reflecting the rapid obsolescence common in high-tech products. Each customer has a unique reservation price, r_k , reflecting their valuation of the AI product at time interval k . This price is modelled as a product of the previous interval's price and a random change rate X_k , capturing the dynamic nature of market conditions and customer perceptions of new technology:

$$r_k = r_{k+1}X_k = r_T \prod_{i=T}^k X_i \text{ for } T > k \geq 1.$$

Here, r_T is the initial launch price, and X_k are independent, identically distributed random variables representing fluctuations in market demand and technological advancements. At the beginning of each time interval k , the seller sets a minimum acceptable price p_k . A sale occurs if the customer's reservation price r_k meets or exceeds p_k , reflecting their willingness to invest in the new AI technology.

The model employs an optimality equation to determine the sequence of minimum acceptable prices that maximize expected profit:

$$V_{k,n}(r_k) = \max\{r_k - c + E[V_{k-1,n-1}(r_{k-1})], E[V_{k-1,n}(r_{k-1})]\} \quad (1)$$

with the boundary conditions $V_{0,\cdot}(\cdot) = 0$ and $V_{\cdot,0}(\cdot) = 0$, where:

- $V_{k,n}(r_k)$ represents the maximum expected profit when there are k time intervals left, n units of the product to sell, and the current reservation price is r_k .
- $r_k - c$ is the immediate profit from selling a unit at price r_k , minus the cost c of the product.
- $E[V_{k-1,n-1}(r_{k-1})]$ is the expected profit for the remaining $k-1$ intervals and $n-1$ units after making a sale at interval k .
- $E[V_{k-1,n}(r_{k-1})]$ is the expected profit if no sale occurs in interval k and the seller moves to the next interval with all n units still available.

4 Analysis

First, we derive several properties of the objective function $V_{k,1}(r_k)$ and the optimal policy for maximizing it. Then, we obtain the optimal policy for $V_{k,2}(r_k)$.

4.1 The Case with Only One Product ($n=1$)

Proposition 1. $V_{k,1}(r)$ is an increasing function of r for a given k . It is an increasing function of k for a given r .

Given an initial reservation price r , to prove $V_{k+1,1}(r) \geq V_{k,1}(r)$, we consider using the minimum acceptable price in the $(m+1)^{\text{th}}$ time interval as the same as the optimal minimum acceptable price of $V_{k,1}(r)$ in the m^{th} time interval from $m = k$ to $m = 1$. Then the expected profit in the situation with $k+1$ time intervals by the first k time intervals is the same as $V_{k,1}(r)$. Moreover, it occurs with a positive probability that the product has not been sold out after k time intervals. If the product had not been sold after k time intervals, there would still be one time interval to sell it and profit more with a positive probability that a customer will come and offer a high price. Thus, the expected profit with $k+1$ time intervals under this special policy is greater than or equal to $V_{k,1}(r)$. Under the optimal decision policy, the expected profit should be at least the same, i.e. $V_{k+1,1}(r) \geq V_{k,1}(r)$. Therefore, we conclude that $V_{k,1}(r)$ increases with k . $\square$

When the initial reservation price is higher, the reservation prices in later time intervals will have the chance to be higher because it is the product of the previous reservation price and an independent random variable. Thus, the expected profit will be higher. When the initial reservation

price is the same, and there are more time intervals, the seller will have more opportunities to sell the product and profit more.

Lemma 1. Suppose $\mu_F \leq 1$. Both $V_{k,1}(r) - r$ and $E[V_{k-1,1}(rX)] - r$ are decreasing in r for any k .

Lemma 2. Suppose $\mu_F \leq 1$. Both $V_{k,1}(r) - r$ and $E[V_{k-1,1}(rX)] - r$ are decreasing in k for any r .

Theorem 1. For a seller with one product, the optimal sale strategy is as follows:

(1) If $\mu_F > 1$, $p_1 = c < p_2 = p_3 \dots = p_k = \dots = +\infty$. In other words, the seller should sell the product as late as possible.

(2) If $\mu_F \leq 1$, there are increasing numbers $p_1 = c \leq p_2 \leq \dots \leq p_k \leq \dots$ such that if there are k remaining time intervals to go and the current reservation price is r_k , the seller should sell the product if $r_k \geq p_k$

For the AI software tool, this model suggests a dynamic pricing strategy that adapts to market conditions and customer responses. The tool's pricing should be adjusted based on predictive analytics, considering the likelihood of future price increases or decreases. This approach ensures maximizing profits while catering to the market's evolving needs and expectations in the realm of AI technology.

According to Theorem 1, if customers are expected to offer higher prices in the future (i.e. $\mu_F > 1$), the seller can expect a higher profit by selling the product later; if customers are expected to offer lower prices in the future, given the price change distribution, the seller would like to set a higher minimum acceptable price if he/she has more time to sell the product. Given the price change rate distribution, it is possible to find the minimum acceptable price. According to Theorem 1, the optimal policy is to sell the product as late as possible if $\mu_F > 1$. Thus, here, we show one example of the case with $\mu_F \leq 1$.

Example 1. The reservation price change rate follows a uniform distribution (L, U) . The following figure shows that the minimum acceptable price at the last second time interval (i.e., p_2) increases

when L increases from 0.50 to 0.77 with an interval of 0.001 for the case where $c=10$, and $U=1.2$. From Theorem 1, $p_1 = c = 10$.

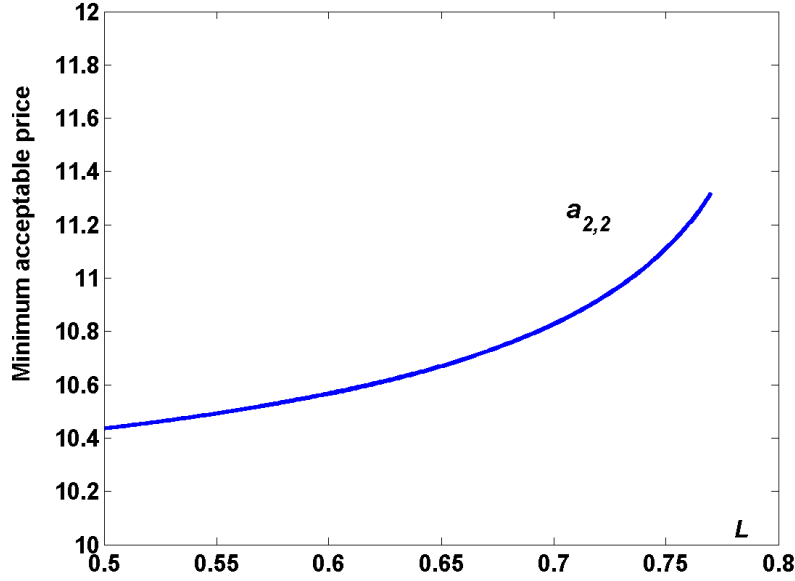


Figure 1. Minimum Acceptable Price At The Last Second Time Interval

Although the lower limit L is less than 1, the upper limit $U=1.2$ is greater than 1. Thus, the seller still has some chance to attract customers whose reservation prices are higher. When the seller has two products, p_1 is the minimum acceptable price of the last product (i.e., after selling one product). p_2 is the minimum acceptable price of the first product. Thus, p_2 is slightly higher than the cost $c=10$.

4.2 The Case with Two Products

Let us now study the optimal policy for selling the first product. Note that after selling one product, the remaining problem is the same as the case in the previous subsection.

Proposition 2. $V_{k,2}(r)$ is increasing in r for any given k and increasing in k for any given r .

To simplify the notation henceforth, we define the following two functions, which are similar to those in Taylor (1967). For $k \geq 2$,

$$H_k(r) \equiv V_{k,2}(r) - r - E[V_{k-1,1}(rX)] \quad (2)$$

$$J_k(r) \equiv E[V_{k-1,2}(rX)] - r - E[V_{k-1,1}(rX)]$$

$$= E[H_{k-1}(rX)] + r(\mu_F - 1) - E[V_{k-1,1}(rX) - E[V_{k-2,1}(rX_k X_{k-1})|X_k]] \quad (3)$$

$$\Rightarrow H_k(r) = \max \{-c, J_k(r)\} \quad (4)$$

In particular, $H_1(r) = \max \{-c, -r\}$ and $J_1(r) = -r$.

Since $V_{k,2}(r) - r - E[V_{k-1,1}(rX)] = \max \{-c, E[V_{k-1,2}(r)] - r - E[V_{k-1,1}(rX)]\}$, if $J_k(r) < -c$, the seller will sell one product at once; if $J_k(r) = -c$, there is no difference between selling and not selling one product to the current customer, and if $J_k(r) > -c$, it is better not to sell one product to the current customer.

Lemma 3. For any k , if $\mu_F \leq 1$, then both $J_k(r)$ and $H_k(r)$ are decreasing in r .

Lemma 4. For any r , if $\mu_F \leq 1$, then both $J_k(r)$ and $H_k(r)$ are increasing in k .

From Lemma 2, we have the following corollary immediately.

Corollary 1. If $\mu_F \leq 1$ and there exists a positive integer z such that $J_{z+1}(r) \geq J_z(r)$ for any r , then if $k \geq z$, $J_k(r)$ and $H_k(r)$ will be increasing in k for any r .

Now we can find the optimal strategy for selling the first product. Let a_k be the minimum acceptable price of the first product when there are k remaining time intervals to go.

Theorem 2. The optimal policy for selling the first product has the following form:

- (1) If $\mu_F > 1$, there are increasing numbers $a_1 = c = a_2 < a_3 = \dots = +\infty$ such that the first product should be sold if $r_k \geq a_k$. In other words, the seller should sell the product as late as possible.
- (2) if $\mu_F \leq 1$, there are increasing numbers $a_1 = c = a_2 \leq \dots \leq a_k \leq \dots$ such that the first product should be sold if $r_k \geq a_k$.

Theorem 2 is similar to Theorem 1. If the expectation of the offered price keeps increasing (i.e., $\mu_F > 1$), the seller can obtain a higher expected profit by selling the products later. If the expectation of the offered price keeps decreasing (i.e., if $\mu_F < 1$), the seller would like to set higher minimum acceptable prices when there are more remaining time intervals to sell products.

5. Two Products with Differentiated Costs

In our AI-driven analysis, we examine a scenario involving two products with distinct cost structures. We denote c_1 and c_2 as the salvage values or costs of the first and second products, respectively, where $c_1 \neq c_2$. In this context, the AI model employs an optimality equation

$V_{k,2}(r_k) = \max\{r_k - c_1 + E[V_{k-1,1}(r_k X)], E[V_{k-1,2}(r_k X)]\}$, which is crucial for determining dynamic pricing strategies under these cost differentials, and $V_{k,1}(r_k) = \max\{r_k - c_2, E[V_{k-1,1}(r_k X)]\}$, with the boundary conditions $V_{\cdot,0}(r) = 0$ and $V_{0,\cdot}(r) = 0$. In particular, $V_{1,2}(r_1) = \max\{r_1 - c_1, 0\}$, and $V_{1,1}(r_1) = \max\{r_1 - c_2, 0\}$.

The second product may be treated the same way as in the previous subsection. Hence, let us focus on the optimal strategy for the first product.

Lemma 5. If $c_1 < c_2$ and $\mu_F \leq 1$, both $J_k(r)$ and $H_k(r)$ are increasing in k for any r .

If $c_1 > c_2$, the following example shows that $J_k(r)$ and $H_k(r)$ may not always increase in k for any r .

Example 2. If $c_1 = 10$, $c_2 = 3$, $P(X=0.7) = 0.9$, $P(X=1.2) = 0.1$, and $r = 10$, then $\mu_F = 0.75 < 1$, $J_1(10) = -10$, $J_2(10) = -14.3$, and $J_3(10) = -10.625$. Thus, $J_1(10) > J_2(10)$, $J_3(10) > J_2(10)$.

If $c_1 = 30$, $c_2 = 3$, $P(X=1.2) = 0.6$, $P(X=0.3) = 0.4$, and $r = 10$, then $\mu_F = 0.84 < 1$, $J_1(10) = -10$, $J_2(10) = -15.4$, and $J_3(10) = -15.76$. Thus, $J_1(10) > J_2(10) > J_3(10)$.

Theorem 3. For selling the first product, the optimal policy has the following form: $a_1 = c_1$; $a_2 \geq c_2$ if $c_2 < c_1$, and $a_2 \leq c_2$ if $c_2 > c_1$. Moreover,

(1) If $\mu_F > 1$, then $a_2 < a_3 = \dots = a_k = \dots = +\infty$. In other words, the seller should sell the product as late as possible.

(2) If $\mu_F \leq 1$ and $c_1 < c_2$, then $a_2 \leq a_3 \leq \dots \leq a_k \leq \dots$.

(3) If $\mu_F \leq 1$ and $c_1 > c_2$, then $a_2, a_3, \dots, a_k, \dots$ may or may not be increasing.

To find a_k when $k \geq 3$, we consider two cases:

Case 1. Suppose that $\mu_F > 1$.

If the seller sells the first product in the k^{th} time interval, where $k > 2$, then he/she can obtain the expected profit $r_k - c_1 + E[V_{k-1,1}(rX_k)]$. However, if the seller sells the first product in the $(k-1)^{\text{th}}$ time interval, then the expected profit becomes $r_2 - c_1 + E[V_{1,1}(r_2X)]$. We have proved that it is better to sell the second product as late as possible. This means that regardless of whether the seller sells the first product at the k^{th} time interval or one time interval later, the profit from the second product depends only on the reservation price at the last time interval. Hence, $E[V_{k-1,1}(rX_k)] = E[V_{k-2,1}(rX_kX_{k-1})]$. Then, $r - c_1 + E[V_{k-1,1}(rX_k)] \leq \mu_F r - c_1 + E[V_{k-2,1}(rX_kX_{k-1})]$. Therefore, it is better to sell the first product as late as possible, i.e., $a_3 = \dots = a_k = \dots = +\infty$.

Case 2. Suppose that $\mu_F \leq 1$. If $c_1 < c_2$, the proof of $a_2 \leq a_3 \leq \dots \leq a_k \leq \dots$ is the same as that of Theorem 2; if $c_1 > c_2$, we consider the following two examples.

(4) $a_3 > a_2$ if $c_1 = 10$, $c_2 = 3$, $P(X=0.7)=0.9$, and $P(X=1.2)=0.1$. We can obtain a_2 by solving the inequality $J_2(r) \leq -10$.

$$\begin{aligned} J_2(r) &\leq -10 \Leftrightarrow E[\max\{rX - 10, 0\}] - E[\max\{rX - 3\}] - r + 10 \leq 0 \\ &\Leftrightarrow \max\{0.7r - 10, 0\} \times 0.9 + \max\{1.2r - 10, 0\} \times 0.1 - \max\{0.7r - 3, 0\} \times 0.9 - \max\{1.2r - 3, 0\} \times 0.1 - r + 10 \leq 0 \end{aligned} \quad (5)$$

(1.1) If $0 \leq r \leq \frac{3}{1.2}$, the inequality (5) becomes $-r + 10 \leq 0$. This is impossible.

(1.2) If $\frac{3}{1.2} < r \leq \frac{3}{0.7}$, the inequality (5) becomes $-0.1 \times (1.2r - 3) - r + 10 \leq 0$, i.e., $r \geq \frac{10.3}{1.12}$.

This is also impossible.

(1.3) If $\frac{3}{0.7} < r \leq \frac{10}{1.2}$, the inequality (5) becomes $-0.9 \times (0.7r - 3) - 0.1 \times (1.2r - 3) - r + 10 \leq 0$, i.e., $r \geq \frac{13}{1.75}$. This is possible.

Therefore, $a_2 = \frac{13}{1.75}$. However, $J_3\left(\frac{13}{1.75}\right) = -0.62346 \leq -10$. By Lemma 3, we have $a_3 > a_2$.

(2) $a_3 < a_2$ if $c_1 = 30$, $c_2 = 3$, $P(X=1.2)=0.6$, and $P(X=0.3)=0.4$. We have similarly

$$J_2(r) \leq -30 \Leftrightarrow E[\max\{rX - 30, 0\}] - E[\max\{rX - 3\}] - r + 30 \leq 0$$

$$\Leftrightarrow \max\{0.3r - 30, 0\} \times 0.4 + \max\{1.2r - 30, 0\} \times 0.6 - \max\{0.3r - 3, 0\} \times 0.4 - \max\{1.2r - 3, 0\} \times 0.6 - r + 30 \leq 0 \quad (6)$$

(2.1) If $0 \leq r \leq \frac{3}{1.2}$, the inequality (6) becomes $-r + 30 \leq 0$. This is impossible.

(2.2) If $\frac{3}{1.2} < r \leq \frac{3}{0.3}$, the inequality (6) becomes $-0.6 \times (1.2r - 3) - r + 30 \leq 0$, i.e., $r \geq \frac{31.8}{1.72}$. This is also impossible.

(2.3) If $\frac{3}{0.3} < r \leq \frac{30}{1.2}$, the inequality (6) becomes $-0.4 \times (0.3r - 3) - 0.6 \times (1.2r - 3) - r + 30 \leq 0$, i.e., $r \geq \frac{33}{1.84}$. This is possible.

Therefore, $a_2 = \frac{33}{1.84}$. However, $J_3\left(\frac{33}{1.84}\right) = -37.0069 < -30$. By Lemma 3, $a_3 < a_2$.

6. Numerical Examples on AI-Driven vs Traditional Pricing

In this part, we use simplified numerical examples to create comparative plots. These examples will highlight how AI-driven dynamic pricing adapts to market changes and optimizes profitability more effectively than traditional methods.

In this example, AI-driven pricing adjusts prices dynamically based on historical sales data, predictive analytics, and remaining sales time, while the traditional method adopts fixed pricing or periodic manual adjustments without real-time data integration. In the AI-driven pricing, we set the initial reservation price $r_T = 100$, cost $c = 10$, time intervals $(T) = 10$, and price change rates (x_k) to follow a uniform distribution between 0.8 and 1.2. For the traditional method, we adopt a fixed price decrease of 2% per time interval. We then create a plot showing how the prices are adjusted over time in both methods. The result is shown in Figure 2.

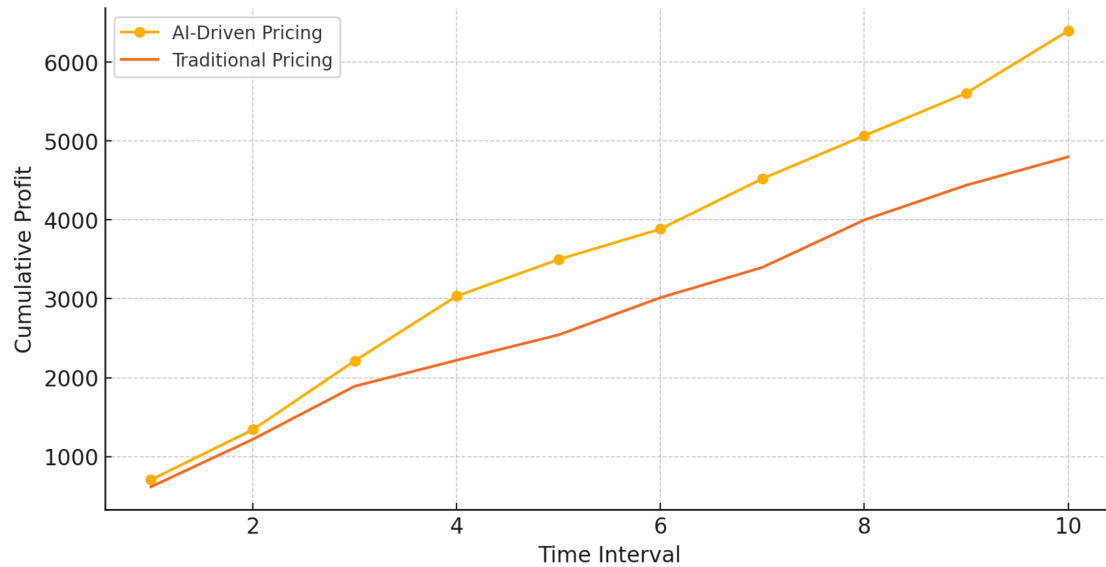


Figure 2. Price Adjustment Over Time: the AI-Driven vs. Traditional methods

Figure 2 shows how prices are adjusted over time using AI-driven pricing compared to traditional pricing. The AI-driven approach results in more dynamic and varied price adjustments, reflecting its responsiveness to market changes and predictive analytics, whereas traditional pricing shows a steady, linear decrease. Next, we create a plot comparing the cumulative profit over time for these two methods.

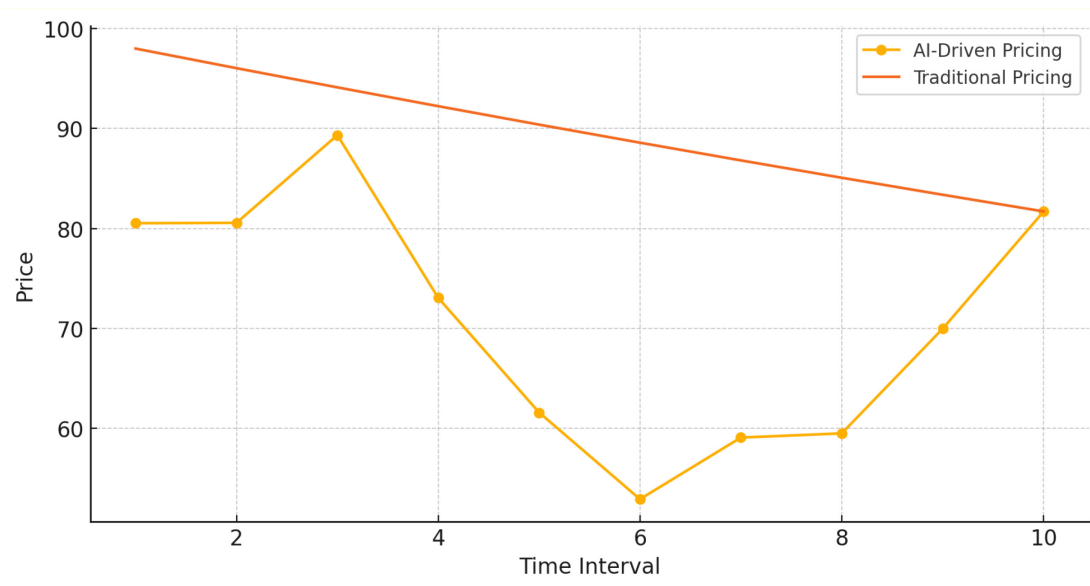


Figure 3. The cumulative Profit Over Time

Figure 3 illustrates that the AI-driven pricing strategy results in a higher cumulative profit compared to the traditional method, demonstrating the superior ability of AI to optimize pricing by adapting to market conditions and maximizing sales revenue. In summary, these figures show that compared to traditional pricing, AI can better analyze large datasets, quickly adapt to market changes, and improve profit.

7. Discussion and Concluding Remarks

This paper offers a strategic framework for AI-enhanced dynamic pricing in the context of selling one or two high-value products within a finite time horizon, particularly in manufacturing and service industries. This approach is crucial in today's market, where firms often face the challenge of optimizing sales opportunities for premium products within a limited timeframe. Traditionally, pricing strategies assumed that each new potential customer's valuation was independent of previous customers. However, with the advent of the internet and social media, potential customers are now more informed about pricing trends, making their valuation dependent on a broader market perspective. This interconnectivity and information flow make AI-driven dynamic pricing both fascinating and complex. Our study innovatively integrates AI to model scenarios where a new customer's valuation is influenced by previous ones, paving the way for future research in AI-based pricing strategies where customer valuation is affected by historical data.

In exploring profit maximization through asymmetric information, we delve into scenarios like retailers displaying products nearing expiration without full inventory disclosure, as discussed by Chung (2019). The integration of AI in inventory management and pricing transparency, as suggested by Tunuguntla et al. (2019), opens new avenues for dynamic pricing strategies. This paper particularly focuses on AI's role in scenarios where sellers disclose competitive pricing information, as explored by Kim et al. (2020). The study extends to more realistic market scenarios with multiple sellers and AI-driven competitive analysis, as seen in the work of Gisches et al. (2021).

7.1 Strengths and Limitations of AI in Dynamic Pricing

The results of this study and the literature indicate several key strengths and limitations of the use of AI in dynamic pricing. In terms of strengths, first, AI-driven pricing can provide personalized pricing according to individual customer data and historical interactions, therefore improving

customer satisfaction and loyalty. Second, AI-driven dynamic pricing enables us to lively analyze and adjust prices based on historical sales data, market trends, and predictive analytics. This allows companies to optimize profit margins by responding promptly to market conditions and consumer behaviors. Third, By adjusting prices based on inventory levels and remaining sales time, AI-driven pricing helps manage inventory levels effectively. Fourth, AI algorithms integrate data from supply chain, marketing, and finance. This cross-functional integration leads to more informed and consistent pricing decisions that align with business objectives and market conditions. Fifth, AI systems can predict market trends and consumer demand, allowing businesses to set prices to maximize expected revenue.

In terms of limitation, first, implementing AI-driven dynamic pricing systems is complex and costly. It also requires significant investment in technology and skilled staff. Second, the effectiveness of AI in dynamic pricing heavily depends on data quality and availability. Inaccurate or incomplete data may lead to suboptimal pricing decisions. Third, customers may feel unfair or worry about information leaking if it's not handled transparently. Fourth, AI algorithms may struggle with extreme market volatility and unforeseen events, therefore leading to inaccurate pricing adjustments. Fifth, various regulations and consumer protection may decrease the flexibility of AI-driven pricing strategies.

7.2 Comparing AI-Driven VS Traditional Dynamic Pricing Strategies

As shown in the research results of this paper and the literature, there are significant differences in data utilization, responsiveness, implementation complexity, ethical considerations, and operational efficiency when comparing AI-driven and traditional dynamic pricing strategies. Several important differences are summarized below.

Table 1: Traditional Dynamic Pricing vs. AI-Driven Pricing

	AI-Driven Dynamic Pricing	Traditional Dynamic Pricing
Data Utilization and Analysis	• Real-time data analysis and predictive analytics. • Multiple data sources integration.	• Historical data analysis. • Limited data source integration.
Pricing Adjustments and Responsiveness	• Continuous, personalized pricing adjustments. • Scenario simulation for optimal pricing.	• Periodic, segmented pricing adjustments.

		• Limited scenario planning capabilities.
Implementation and Complexity	• Complex implementation. • High initial costs but offers long-term benefits. • Continuous learning • High predictive power.	• Easier implementation. • Less costly to implement • Lack adaptability. • Low predictive power.
Ethical Considerations and Fairness	• Raises transparency and fairness concerns. • More challenges in regulatory compliance and consumer perception.	• More transparent. • Easier to comply with regulations and foster greater consumer trust.
Operational Efficiency	• Automating the pricing process. • Increasing efficiency and scalability.	• Requiring significant manual effort. • Time consuming and less efficient.

7.3 AI-Driven Dynamic Pricing Applications

Lufthansa, Germany's largest airline, uses complex AI algorithms provided by PROS and integrates IBM Watson products to manage, search, analyze, and interpret the various relevant and connected data sources such as booking patterns, historical flight occupancy rates, competitor fares, and seasonal trends and adjust ticket prices in real-time based on demand forecasts and market conditions since the middle of 2022 (Clemente, 2020). The AI system continuously monitors and adjusts ticket prices to optimize revenue. Lufthansa reported revenue increases by 15%, profit more than double, and passenger numbers increase by 20% in 2023. (Lufthansa Group, 2023)

Lufthansa's AI provider PROS, initials user experiences, processes automation, and personalizes purchasing experiences designed to meet the real-time demands for extensive types of industries in Manufacturing, Services, Energy, Technology, Distribution, Transportation, and so on have reported significant improvements in revenue and performances. For example, with PROS AI Powered pricing, Manitou Group, a French heavy equipment manufacturer, delivered over 40,000 orders in 2022 and increased sales by 26% year over year with the PROS AI Price system. McKesson, a leading healthcare services company, has seen preferred suppliers growing three times faster, improved cost and payment terms with suppliers, and accelerated its realization of

profit synergies. Process monitoring manufacturer expands margins by over \$5M within a month of implementation (PROS. n.d.).

When considering multiple products, AI can help analyze a variety of factors. Neutral AI-based pricing models, considering inventory and time constraints as suggested by Katehakis et al. (2019), are explored. The impact of product quality on pricing, a factor increasingly analyzed through AI algorithms, is also considered, following insights from Akçay and Karaesmen (2021). Yang et al. (2022) highlight how AI can adjust pricing strategies when quality is a variable. The categorization of goods and the dependency of prices on initial pricing and product categories, as discussed by Namin et al. (2022), are also integrated into AI models. AI plays a crucial role in understanding the impact of salvage value variations on pricing decisions, as noted by Biswas and Avittathur (2018). In service industries, where the salvage value often drops to zero, AI can optimize pricing strategies within a finite time, as explored by Schur et al. (2019). The use of AI in analyzing historical sales data can help mitigate sellers' regrets and refine future pricing strategies, a concept examined by Li and Liu (2022). Finally, the potential of AI in facilitating advanced selling strategies, benefiting both sellers and customers, is highlighted following the research of Li and Li (2022).

Appendix

Proof of Proposition 1:

We prove the first part by induction on k .

When $k = 1$, $V_{1,1}(r) = \max\{r - c, 0\}$ increases as r increases. Now suppose that $V_{k,1}(r)$ is an increasing function of r if $k = m$. If $k = m + 1$,

$$V_{m+1,1}(r) = \max\{r - c, E[V_{m,1}(rX)]\}.$$

$E[V_{m,1}(rX)]$ is an increasing function of r because $V_{m,1}(rX)$ is an increasing function of r for each value of X . Thus, $V_{m+1,1}(r)$ is an increasing function of r . By induction, we conclude that $V_{k,1}(r)$ is an increasing function of r in general.

Proof of Lemma 1:

We prove the conclusion by induction on k . $V_{k,1}(r) - r = \max\{-c, E[V_{k-1,1}(rX)] - r\}$. When $k=1$, $V_{1,1}(r) - r = \max\{-c, -r\}$ is decreasing in r . Now, let m be a given positive integer, and suppose that $V_{k,1}(r) - r$ is decreasing in r when $k \leq m$. Then, for $k = m+1$,

$$V_{m+1,1}(r) - r = \max\{-c, E[V_{m,1}(rX)] - r\} = \max\{-c, E[V_{m,1}(rX) - rX] + r(\mu_F - 1)\}.$$

For each given X , $V_{m,1}(rX) - rX$ is decreasing in r , and $r(\mu_F - 1)$ is decreasing in r since $\mu_F \leq 1$. Thus, $V_{m+1,1}(r) - r$ is decreasing in r . Hence, we conclude that $V_{k,1}(r) - r$ is decreasing in r for each k .

Next, we can prove that $E[V_{k-1,1}(rX)] - r$ is decreasing in r since $E[V_{k-1,1}(rX)] - r = E[V_{k-1,1}(rX) - rX] + r(\mu_F - 1)$.

Proof of Lemma 2:

We prove the conclusion by induction on k . $E[V_{0,1}(rX)] - r = -r$; $E[V_{1,1}(rX)] - r \geq -r$ since $V_{1,1}(rX) \geq 0$ for any rX . Thus, $E[V_{1,1}(rX)] - r \geq E[V_{0,1}(rX)] - r$. Also, we have $V_{1,1}(r) - r = \max\{-c, -r\}$ for any r ; $E[V_{2,1}(r)] - r = \max\{-c, E[V_{1,1}(rX)] - r\} \geq \max\{-c, -r\} = V_{1,1}(r) - r$ for any r since $V_{1,1}(rX) \geq 0$ for any rX .

Now, let m be a given positive integer, and suppose that $E[V_{k-1,1}(rX)] - r \geq E[V_{k-2,1}(rX)] - r$ and $E[V_{k,1}(r)] - r \geq E[V_{k-1,1}(r)] - r$ for any r when $k \leq m$. Then, for $k = m+1$,

$$E[V_{m,1}(r)] - rX = \int [V_{m,1}(rt) - rt] dF(t) \geq \int [V_{m-1,1}(rt) - rt] dF(t) = E[V_{m,1}(rX) - rX].$$

On the other hand, we know that in general, $V_{k,1}(r) - r = \max\{-c, E[V_{k-1,1}(rX)] - r\}$. Hence,

$$V_{m+1,1}(r) - r \geq V_{m,1}(r) - r.$$

Proof of Theorem 1:

(1) Suppose that $\mu_F > 1$.

When $k=1$, $V_{1,1}(r) = \max\{r_1 - c, 0\}$. Thus, the seller should sell the product if $r_1 \geq c$, i.e., $p_1 = c$. For any $k \geq 2$, if the seller sells the product in the k^{th} time interval, the profit is given by $r_k - c$; however, if the seller delays by one time interval to sell the product, the profit becomes

$\mu_F r_k - c$, which is greater than $r_k - c$ since $\mu_F > 1$. This means that it is better to sell the product as late as possible, i.e. $p_2 = p_3 \dots = p_k = \dots = +\infty$.

(2) Suppose that $\mu_F \leq 1$.

When $k=1$, $V_{1,1}(r) = \max \{r_1 - c, 0\}$. Thus, the seller should sell the product if $r_1 \geq c$, i.e., $p_1 = c$. For any $k \geq 2$, let $p_k = \min \{r: E[V_{k-1,1}(rX)] - r = -c\}$ where we take $p_k = +\infty$ if this set is empty. From Lemma 1, for all $r \geq p_k$, we have $E[V_{k-1,1}(rX)] - r \leq -c$ or $E[V_{k-1,1}(rX)] \leq r - c$, which shows that it is optimal to sell the product if there are k remaining time intervals, and the customer offers the price more than p_k since $V_{k,1}(r) = \max \{r - c, E[V_{k-1,1}(rX)]\}$.

According to the definition of p_{k-1} , $E[V_{k-1,1}(p_{k-1}X)] - p_{k-1} = -c = E[V_{k,1}(p_kX)] - p_k$.

By Lemma 2, we have $E[V_{k-1,1}(p_{k-1}X)] - p_{k-1} \leq E[V_{k,1}(p_kX)] - p_k$. Thus,

$E[V_{k-1,1}(p_kX)] - p_k \leq E[V_{k-1,1}(p_{k-1}X)] - p_{k-1}$. By Lemma 1, $p_k \geq p_{k-1}$. This completes the proof of (2).

Proof of Proposition 2:

(1) We prove the first part by induction on k . Obviously, $V_{k,2}(r) = \max \{r - c, 0\}$ is increasing in r . Let m be a given positive integer, and suppose that $V_{k-1,2}(r)$ is increasing in r when $k \leq m$. Now, for $k=m+1$, $V_{k,2}(r) = \max \{r - c + E[V_{k-1,1}(rX)], E[V_{k-1,2}(rX)]\}$. By the induction hypothesis, $V_{k-1,2}(rX)$ is increasing in r for each X , and $V_{k-1,1}(rX)$ has been proved to be increasing in r for each X in Proposition 1. Therefore $V_{k,2}(r)$ is increasing in r .

(2) Here, we need to prove $V_{k,2}(r) \geq V_{k-1,2}(r)$.

If there are $k-1$ remaining time intervals and the reservation price is r , let g be the optimal policy to obtain $V_{k-1,2}(r)$.

Now, assume there are k remaining time intervals and the reservation price is r . We apply the same policy g on the first $k-1$ time intervals. Then, both situations (with k or $k-1$ remaining time intervals) have the same reservation prices and the same minimum acceptable prices on the first $k-1$ time intervals. Therefore, the expected profit in the situation with k remaining time intervals

during the first $k-1$ time intervals is just $V_{k-1,2}(r)$. Regarding the last time interval, there are two possibilities:

(2.1) Two products have been sold during the first $k-1$ time intervals. The expected profit in the situation with k remaining time intervals under the special policy g is the same as $V_{k-1,2}(r)$.

(2.2) One or two products have not been sold during the first $k-1$ time intervals. In the last time interval, the reservation price can be greater than c with a positive probability. This means that one product can be sold to profit more with a positive probability.

Hence, the expected profit in the situation with k remaining time interval under the special policy g is greater than $V_{k-1,2}(r)$. Hence, under the optimal policy, the expected profit should be at least the same, i.e. $V_{k,2}(r) \geq V_{k-1,2}(r)$.

Proof of Lemma 3:

We prove it by induction on k . When $k=1$, the conclusion is clear since $H_1(r) = \max\{-c, -r\}$ and $J_1(r) = -r$.

When $k = 2$, we have

$$\begin{aligned} J_2(r) &= E[V_{1,2}(rX)] - r - E[V_{1,1}(rX)] \\ &= E[V_{1,2}(rX) - rX] + r(\mu_F - 1) - E[V_{1,1}(rX)], \end{aligned}$$

which is decreasing in r because the three terms on the right of the last equation $E[V_{1,2}(rX) - rX]$, $r(\mu_F - 1)$, and $-E[V_{1,1}(rX)]$ are decreasing in r . From equation (3), we know that $H_2(r)$ is decreasing in r too.

Assume $J_{k-1}(r)$ and $H_{k-1}(r)$ are decreasing in r . Then, $E[H_{k-1}(rX)]$ is decreasing in r . To simplify the notation, we write $V_{k-1,1}(rX) - E[V_{k-2,1}(rX_k X_{k-1})|X_k]$ as $V_{k-1,1}(r_{k-1}) - E[V_{k-2,1}(r_{k-1}X)]$. As we know, $V_{k-1,1}(r_{k-1}) = \max\{r_{k-1} - c, E[V_{k-2,1}(r_{k-1}X)]\}$. If $r_{k-1} < p_{k-1}$, $V_{k-1,1}(r_{k-1}) = E[V_{k-2,1}(r_{k-1}X)]$; if $r_{k-1} \geq p_{k-1}$, $V_{k-1,1}(r_{k-1}) - E[V_{k-2,1}(r_{k-1}X)] = r_{k-1} - c - E[V_{k-2,1}(r_{k-1}X)] = r_{k-1}(1 - \mu_F) - c - E[V_{k-2,1}(r_{k-1}X) - r_{k-1}X]$, which is increasing in r_{k-1} by Lemma 1. Therefore, $-E[V_{k-1,1}(rX) - E[V_{k-2,1}(rX_k X_{k-1})|X_k]]$ is decreasing in r .

It follows that $J_k(r)$ is decreasing in r because the three terms on the right of Equation (3) $E[H_{k-1}(rX)]$, $r(\mu_F - 1)$, and $-E[V_{k-1,1}(rX) - E[V_{k-2,1}(rX_kX_{k-1})|X_k]]$ are decreasing in r . Hence, Equation (4) shows that $H_k(r)$ is decreasing in r . The proof is complete.

Proof of Lemma 4:

We prove it by induction on k . By definition, we know $J_2(r) = J_1(r) = -r$, and $H_2(r) = \max\{-c, -r\} = H_1(r)$ for any r .

Let m be a given positive integer, and suppose that $J_k(r) \geq J_{k-1}(r)$ and $H_k(r) \geq H_{k-1}(r)$ for any r when $k \geq m$. When $k = m+1$, we study the two terms $E[H_{k-1}(rX)]$, and $-E[V_{k-1,1}(rX) - E[V_{k-2,1}(rX_kX_{k-1})|X_k]]$ of $J_k(r)$ in Equation (2):

(1) For the term $E[H_{k-1}(rX)]$, we have

$$E[H_m(rX)] = \int H_m(rt)dF(t) \geq \int H_{m-1}(rt)dF(t) = E[H_{m-1}(rX)].$$

2) For the term $-E[V_{k-1,1}(rX) - E[V_{k-2,1}(rX_kX_{k-1})|X_k]]$, we have

$$V_{m,1}(r_m) - E[H_{m-1,1}(r_mX)] = \max\{r_m - c - E[H_{m-1,1}(r_mX)], 0\}$$

$$V_{m-1,1}(r_m) - E[H_{m-2,1}(r_mX)] = \max\{r_m - c - E[H_{m-2,1}(r_mX)], 0\}$$

Since $E[H_{m-1,1}(r_mX)] \geq E[H_{m-2,1}(r_mX)]$ according to Proposition 1, we have

$$V_{m,1}(r_m) - E[H_{m-1,1}(r_mX)] \leq V_{m-1,1}(r_m) - E[H_{m-2,1}(r_mX)].$$

Thus, $-E[V_{m,1}(rX_{m-1}) - E[V_{m-1,1}(rX_{m+1}X_m)|X_{m+1}]] \geq -E[V_{m-1,1}(rX_m) - E[V_{m-2,1}(rX_mX_{m-1})|X_m]]$.

Hence, $J_{m+1}(r) \geq J_m(r)$ and so $H_{m+1}(r) \geq H_m(r)$ for any r .

Proof of Theorem 2:

From $V_{1,2}(r) = \max\{r - c, 0\}$, we have $a_1 = c$.

When $k=2$, if the seller sells the first product, he/she can obtain the profit $r_2 - c + E[V_{1,1}(r_2X)]$; if the seller delays one time interval to sell the first product, the profit becomes

$E[V_{1,2}(r_2X)]$ since there will be only one time interval for selling one product. Moreover, $E[V_{1,1}(r_2X)] = E[V_{1,2}(r_2X)]$. Thus, $a_2 = c$.

(1) Suppose that $\mu_F > 1$. If $k > 2$ and if the seller sells the first product in the k^{th} time interval, he/she can obtain the expected profit $r_k - c + E[V_{k-1,1}(r_kX_k)]$; however, if the seller delays one time interval (i.e., in the $(k-1)^{\text{th}}$ time interval) to sell the first product, the expected profit becomes $\mu_F r_k - c + E[V_{k-2,1}(r_kX_kX_{k-1})]$. On the other hand, we have proved that if $\mu_F > 1$, it is better to sell the second product as late as possible. This means that regardless of whether the seller sells the first product in the k^{th} time interval or the following time interval, the profit from the second product depends only on the reservation price in the last time interval. Hence, $E[V_{k-1,1}(r_kX_k)] = E[V_{k-2,1}(r_kX_kX_{k-1})]$. We have $r_k - c + E[V_{k-1,1}(r_kX_k)] < \mu_F r_k - c + E[V_{k-2,1}(r_kX_kX_{k-1})]$ since $\mu_F > 1$. This means that it is better to sell the first product as late as possible, i.e. $a_3 = \dots a_k = \dots = +\infty$.

(2) Suppose that $\mu_F \leq 1$. If $k > 2$, let have $a_k = \min\{r: J_k(r) = -c\}$ where we take $a_k = +\infty$ if this set is empty. By Lemma 3, for all $r \geq a_k$, we have $J_k(r) \leq -c$, thus showing that it is optimal to sell the first product when the reservation state is r_k if $r_k \geq a_k$.

Now, $J_{k-1}(a_{k-1}) = -c$, and $J_k(a_k) = -c$. By Lemma 2, we have $J_{k-1}(a_k) \leq J_k(a_k)$. Thus, $J_{k-1}(a_k) \leq J_{k-1}(a_{k-1})$. By Lemma 1, $a_k \geq a_{k-1}$.

Proof of Lemma 5:

Almost the same as Lemma 4. So it is omitted here.

Proof of Theorem 3:

Clearly, $a_1 = c_1$.

In particular, in the last second time interval (i.e., $k = 2$), if the seller sells the first product, he/she can obtain the profit $r_2 - c_1 + E[V_{1,1}(r_2X)]$; however, if the seller does not sell the first product, the profit becomes $E[V_{1,2}(r_2X)]$. To determine a_2 , Let us start with the relationship

$$r_2 - c_1 + E[V_{1,1}(r_2X)] \geq E[V_{1,2}(r_2X)] \Leftrightarrow \int_{r_2}^{+\infty} (r_2t - c_2)dF(t) \geq \int_{r_2}^{+\infty} (r_2t - c_1)dF(t).$$

If $c_2 < c_1$,

$$r_2 - c_1 + \int_{\frac{c_2}{r_2}}^{+\infty} (r_2 t - c_2) dF(t) \geq \int_{\frac{c_1}{r_2}}^{+\infty} (r_2 t - c_1) dF(t) \Leftrightarrow r_2 - r_2 \int_{\frac{c_1}{r_2}}^{+\infty} F(t) df \geq c_2.$$

Hence, $a_2 \geq c_2$. If $c_2 > c_1$, we have:

$$r_2 - c_1 + \int_{\frac{c_2}{r_2}}^{+\infty} (r_2 t - c_2) dF(t) \geq \int_{\frac{c_1}{r_2}}^{+\infty} (r_2 t - c_1) dF(t) \Leftrightarrow r_2 + r_2 \int_{\frac{c_1}{r_2}}^{\frac{c_2}{r_2}} F(t) df \geq c_2.$$

Hence, $a_2 \leq c_2$.

References

- Ahn, H.S., M. Gümüs, and P., Kaminsky. 2007. Pricing and Manufacturing Decisions When Demand Is a Function of Prices in Multiple Periods. *Operations Research*, 55(6), 1039-1057.
- Akçay, Y., and F., Karaesmen. 2021. Customers know best: Pricing policies for products with heterogeneous quality. *Decision Sciences*, 1-12.
- Akilandeswari, S. V., Nagpal, P., Vinotha, C., Jain, K., Chatterjee, R., & Gundavarapu, M. R. (2024). Transforming E-Commerce: Unleashing The Potential Of Dynamic Pricing Optimization Through Artificial Intelligence For Strategic Management. *Migration Letters*, 21(S3), 1250-1260.
- Araman, V.F., and R. Caldentey. 2009. Dynamic Pricing for Nonperishable Products with Demand Learning. *Operations Research*, 57(5), 1169-1188.
- Besbes, O., and D. Sauré. 2014. Dynamic Pricing Strategies in the Presence of Demand Shifts. *Manufacturing & Service Operations Management*, 16(4), 513-528.
- Biswas, I., and B., Avittathur. 2018. The price-setting limited clearance sale inventory model. *Annals of Operations Research*, 1-17. Doi:10.1007/s10479-018-2811-5.
- Bitran, G.R., and S.V., Mondschein. 1993. Pricing Perishable Products: an Application to the Retail Industry. MIT Working Paper WP#3592-93, Cambridge, MA, July.
- Bitran, G.R., and S.V., Mondschein. 1997. Periodic Pricing of Seasonal Products in Retailing. *Management Science*, 43(1), 64-79.
- Castañé, G., Dolgui, A., Kousi, N., Meyers, B., Thevenin, S., Vyhmeister E., & Östberg, P-O. (2023) The ASSISTANT project: AI for high level decisions in manufacturing. *International Journal of Production Research*, 61(7), 2288-2306.
- Chatwin, R.E. 2000. Optimal dynamic pricing of perishable products with stochastic demand and a finite set of prices. *European Journal of Operational Research*, 125(1), 149-174.
- Chen, Q., S., Jasin, and I., Duenyas. 2015. Real-Time Dynamic Pricing with Minimal and Flexible Price Adjustment. *Management Science*, 62(8), 2437-2455.
- Chen, X., P., Hu, and Z., Hu. 2016. Efficient Algorithms for the Dynamic Pricing Problem with Reference Price Effect. *Management Science*, 63(12):4389-4408.
- Chen, Y.W., and V.F., Farias. 2013. Simple Policies for Dynamic Pricing with Imperfect Forecasts. *Operations Research*, 61(3), 612-624.

- Chew, EP, C.L., Lee, and R.J., Liu. 2009. Joint inventory allocation and pricing decisions for perishable products. *International Journal of Production Economics*, 120(1), 139-150.
- Chien, C. F., Dauzère-Pérès, S., Huh, W. T., Jang, Y. J., & Morrison, J. R. (2020). Artificial intelligence in manufacturing and logistics systems: algorithms, applications, and case studies. *International Journal of Production Research*, 58(9), 2730-2731.
- Chung, J. 2019. Effective Pricing of Perishables for a More Sustainable Retail Food Market. *Sustainability*, 11(17), 4762.
- Clemente, J. (2020, July 8). Scaling AI at Lufthansa: Combined talents help the airline raise efficiency. IBM Blog. <https://www.ibm.com/blog/scaling-ai-at-lufthansa/>
- De Giovanni, P. (2021). Smart Supply Chains with vendor managed inventory, coordination, and environmental performance. *European Journal of Operational Research*, 292(2), 515-531.
- Feng, Y.Y., and G., Gallego. 1995. Optimal Starting Times for End-of-Season Sales and Optimal Stopping Times for Promotional Fares. *Management Science*, 41(8), 1371-1391.
- Feng, Y.Y., and B.C., Xiao. 2000. A Continuous-Time Yield Management Model with Multiple Prices and Reversible Prices Changes. *Management Science*, 46(5), 644-657.
- Feng, Y.Y., and B.C., Xiao. 2000. Optimal Policies of Yield Management Model with Multiple Predetermined Prices. *Operations Research*, 48(2), 332-343.
- Gallego, G., and G., Van Ryzin. 1994. Optimal Dynamic Pricing of Inventories with Stochastic Demand over Finite Horizons. *Management Science*, 40(8), 999-1020.
- Gallego, G., and M., Hu. 2014. Dynamic Pricing of Perishable Assets under Competition. *Management Science*, 60(5), 1241-1259.
- Gisches, E.J., H., Qi, W.J., Becker, and A., Rapoport. 2021. Strategic retailers and myopic consumers: Competitive pricing of perishable goods. *Journal of Behavioral and Experimental Economics*, 92, 101700
- Güler, M.G., T., Bilgic, and R., Güllü. 2014. Joint inventory and pricing decisions with reference effects. *IIE Transactions*, 46(4), 330-343.
- Hall, J. M., Kopalle, P. K., & Pyke, D. F. (2009). Static and dynamic pricing of excess capacity in a make-to-order environment. *Production and Operations Management*, 18(4), 411-425.
- Katehakis, M.N., Y.F., Liu, and J., Yang. 2019. A revisit to the markup practice of irreversible dynamic pricing. *Annals of Operations Research*, <https://doi.org/10.1007/s10479-019-03438-1>.
- Kim, S., S.M., Slutsky, and B., Thapa. 2020. Optimal Information-Sharing Behaviors among Hotels: Game-Theoretical Approach. *Journal of Hospitality & Tourism Research*, 45(2), 283-303.
- Khosrowabadi, N., Hoberg, K., & Imdahl, C. (2022). Evaluating human behaviour in response to AI recommendations for judgemental forecasting. *European Journal of Operational Research*, 303(3), 1151-1167.
- Li, H.M., and W.T. Huh. 2012. Optimal pricing for a short life-cycle product when customer price-sensitivity varies over time. *Naval Research Logistics*, 59(7), 552-576.
- Li, K.J., and X., Li. 2022. Advance Selling in Marketing Channels. *Journal of Marketing Research*. Accepted.

- Li, M., and Y., Liu. 2022. Managerial Regret and Inventory Pricing. *Management Science*, 68(6), 4398-4414.
- Link, M., Dukino, C., Ganz, W., Hamann, K., & Schnalzer, K. (2020). The Use of AI-Based Assistance Systems in the Service Sector: opportunities, challenges and applications. In *Advances in Human Factors and Systems Interaction: Proceedings of the AHFE 2020 Virtual Conference on Human Factors and Systems Interaction*, July 16-20, 2020, USA (pp. 10-16). Springer International Publishing.
- Liu, Q., and GJ van Ryzin. 2008. Strategic Capacity Rationing to Induce Early Purchases. *Management Science*, 54(6), 1115-1131.
- Lufthansa Group (2023, March 3). Lufthansa Group generates operating profit of 2.7 billion euros in 2023 and invests more than ever for its customers. Lufthansa Group Newsroom. <https://newsroom.lufthansagroup.com/en/lufthansa-group-generates-operating-profit-of-27-billion-euros-in-2023-and-invests-more-than-ever-for-its-customers/>
- Marr, B. (2020). *The intelligence revolution: transforming your business with AI*. Kogan Page Publishers.
- Meredith, J.R., and U., Akinc. 2012. Dynamic Pricing and Demand Estimation for a Single Unique Asset. *Decision Sciences*, 43(4), 711-728.
- Moon, I. E.A. Silver. 2000. The multi-item newsvendor problem with a budget constraint and fixed ordering costs. *Journal of the Operational Research Society*, 51(5), 602-608.
- Namin, A., G.P., Soysal, and B.T., Ratchford. 2022. Alleviating demand uncertainty for seasonal goods: An analysis of attribute-based markdown policy for fashion retailers. *Journal of Business Research*, 145, 671-681.
- Pan, C., Pettit, N.C., Sivanathan, N., Blader, S.L. 2014. Low-status aversion: the effect of self-threat on willingness to buy and sell. *Journal of Applied Social Psychology*, 44, 708-716.
- Petruzzi, N.C., and G.E., Monahan. 2003. Managing Fashion Goods Inventories: Dynamic Recourse for Retailers with Outlet Stores. *IIE Transactions*, 35(11), 1033-1047.
- PROS. (n.d.). Reports on AI-driven dynamic pricing. Retrieved from <https://pros.com/>
- Reim, W., Åström, J., & Eriksson, O. (2020). Implementation of artificial intelligence (AI): a roadmap for business model innovation. *AI*, 1(2), 11.
- Schur, R., J., Gönsch, and M., Hassler. 2019. Time-consistent, risk-averse dynamic pricing. *European Journal of Operational Research*, 277(2), 587-603.
- Seele, P., Dierksmeier, C., Hofstetter, R. and Schultz, M.D., 2021. Mapping the ethicality of algorithmic pricing: A review of dynamic and personalized pricing. *Journal of Business Ethics*, 170, 697-719.
- Sharma, R., Shishodia, A., Gunasekaran, A., Min, H., & Munim, Z. H. (2022). The role of artificial intelligence in supply chain management: mapping the territory. *International Journal of Production Research*, 60(24), 7527-7550.
- Su, X.M. 2007. Intertemporal Pricing with Strategic Customer Behavior. *Management Science*, 53(5), 726-741.

- Swami, S., and P.J., Khairnar. 2006. Optimal normative policies for marketing of products with limited availability. *Annals of Operations Research*, 143(1), 107-121.
- Taylor, H.M. 1967. Evaluating a Call Option and Optimal Timing Strategy in the Stock Market, *Management Science*, 14, 111-120.
- Tunuguntla, V., P., Basu, K., Rakshit, and D., Ghosh. 2019. *European Journal of Operational Research*, 276(1), 119-132.
- Vulcano, G. G.V. Ryzin, and C. Maglaras. 2002. Optimal Dynamic Auctions for Revenue Management. *Management Science*, 48(11), 1388-1407.
- Wansink, B., Kent, RJ, Hoch, SJ. (1998). An anchoring and adjustment model of purchase quantity decisions. *Journal of Marketing Research*, 35(1):71-81.
- Wilson, Timothy D., Christopher E. Houston, Kathryn M. Etling, and Nancy Brekke (1996), "A New Look at Anchoring Effects: Basic Anchoring and Its Antecedents," *Journal of Experimental Psychology: General*, 125 (4), 387-403.
- Xu, X., Hong, Z., Chen, Y., & Cheng, T. C. E. (2023). When is it wise to use artificial intelligence for platform operations considering consumer returns? *European Journal of Operational Research*, 308(3), 1188-1205.
- Yang C.Y., Y.H., Feng, and A. Whinston. 2022. Dynamic Pricing and Information Disclosure for Fresh Produce: An Artificial Intelligence Approach. *Production and Operations Management*, 31(1), 155-171
- Zhao, W., and Y.S., Zheng. 2000. Optimal Dynamic Pricing for Perishable Assets with Nonhomogenous Demand. *Management Science*, 46(3), 375-388.